

Exploratory Data Analysis (EDA) Report

This report analyzes the Superstore dataset to identify sales trends, customer behavior, product performance, and regional profitability, providing insights that support better business decisions.

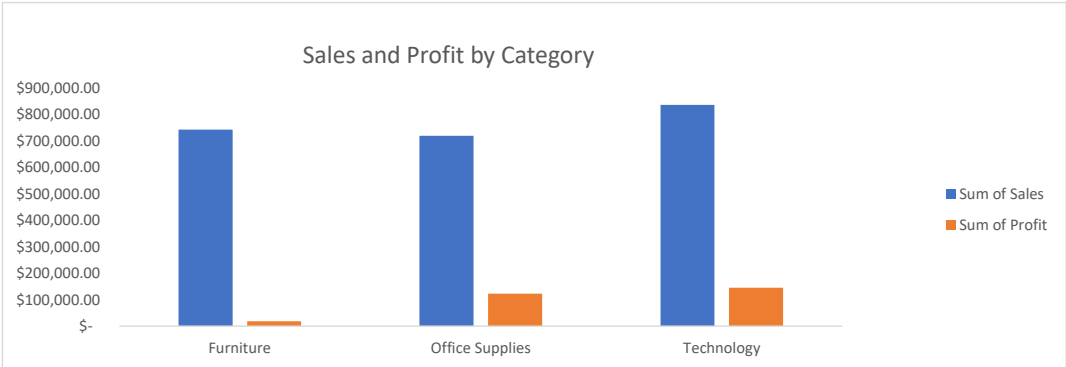
Overall Business Performance

KPI Summary

KPI	Value
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Total Orders	5009
Total Customers	360
Categories	3
Regions	4
Customer Segments	3

Category Performance Analysis

Row Labels	Sum of Sales	Sum of Profit
Furniture	\$ 741,999.80	\$ 18,451.27
Office Supplies	\$ 719,047.03	\$ 122,490.80
Technology	\$ 836,154.03	\$ 145,454.95
Grand Total	2297200.86	286397.0217



Insights

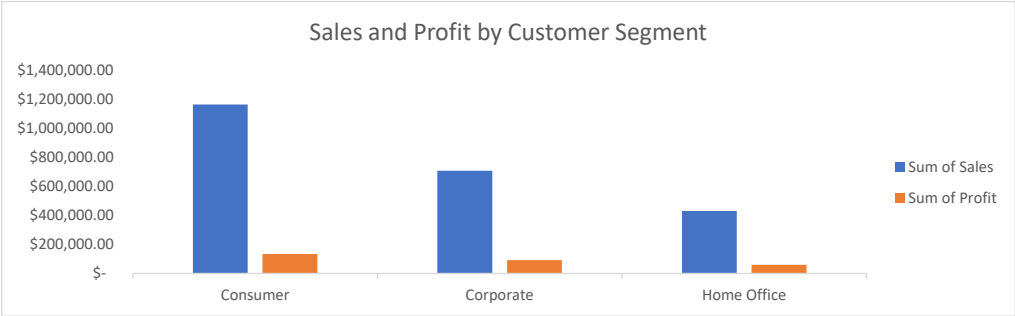
- Technology generated the highest sales and the highest profit, making it the strongest-performing category.
- Office Supplies produced strong sales and healthy profitability, demonstrating consistent performance.
- Furniture generated substantial sales but significantly lower profit than the other categories, suggesting lower profit margins.

Recommendation

Although Furniture contributes considerable revenue, its low profitability indicates margin challenges. Management should investigate product costs, pricing strategies, and discount policies within this category rather than reducing investment altogether. Technology should continue to receive strategic investment, while Office Supplies should be maintained as a stable profit contributor.

Customer Segment Analysis

Row Labels	Sum of Sales	Sum of Profit
Consumer	\$ 1,161,401.34	\$ 134,119.21
Corporate	\$ 706,146.37	\$ 91,979.13
Home Office	\$ 429,653.15	\$ 60,298.68
Grand Total	2297200.86	286397.0217



Insights

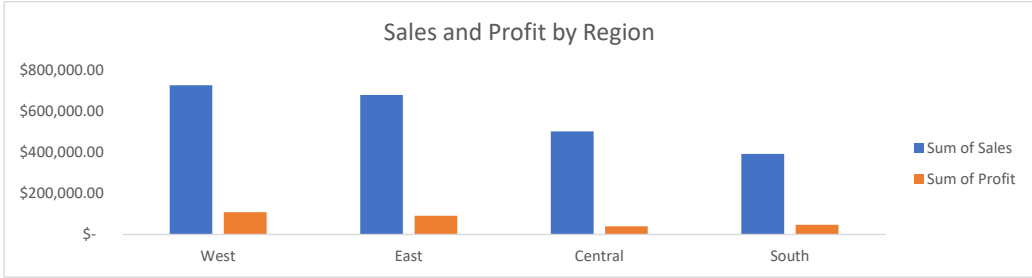
- The Consumer segment generated the highest sales and the highest overall profit, making it the company's largest revenue contributor.
- The Corporate segment recorded lower sales than the Consumer segment but delivered a relatively strong profit, indicating higher profitability per dollar of sales.
- The Home Office segment contributed the lowest sales and profit among the three customer segments.
- No customer segment generated high sales while consistently producing low or negative profit, indicating healthy profitability across all segments.

Recommendation

Continue strengthening the Consumer segment through customer retention and targeted marketing, as it contributes the largest share of revenue and profit. Expand sales efforts toward the Corporate segment because of its strong profitability potential. Meanwhile, investigate opportunities to increase sales within the Home Office segment through targeted promotions and product offerings.

Regional Performance Analysis

Row Labels	Sum of Sales		Sum of Profit	
West	\$	725,457.82	\$	108,418.45
East	\$	678,781.24	\$	91,522.78
Central	\$	501,239.89	\$	39,706.36
South	\$	391,721.91	\$	46,749.43
Grand Total		2297200.86		286397.0217



Insights

- The West region generated the highest sales and profit, making it the strongest-performing region and a key contributor to the company's overall success.
- The East region recorded the second-highest sales and maintained strong profitability, demonstrating stable business performance.
- The Central region generated relatively high sales but significantly lower profit than the West and East, suggesting reduced profit margins.
- The South region recorded the lowest sales and profit, indicating opportunities for market expansion.

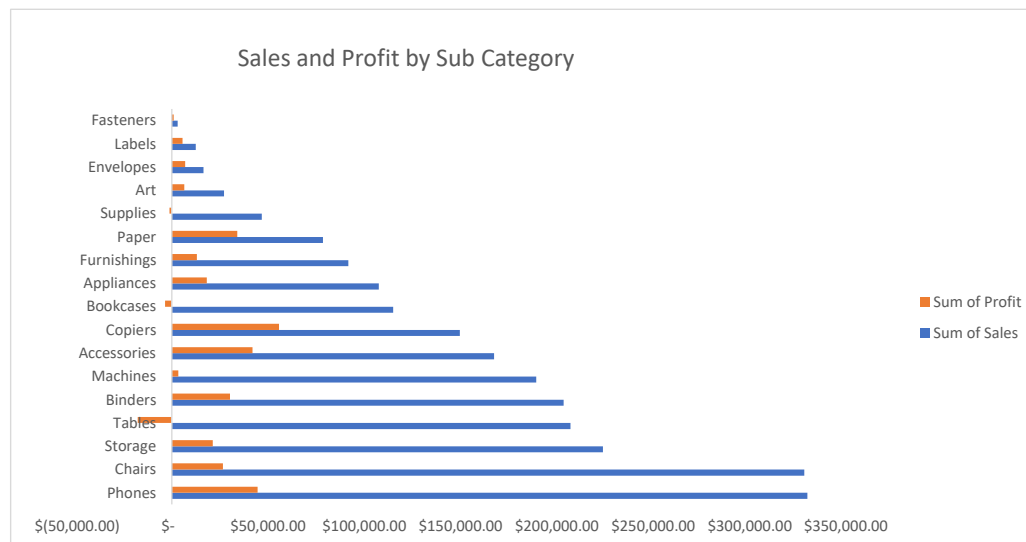
Recommendation

Continue investing in the West region to maintain its market leadership while sustaining growth in the East region. Conduct a detailed investigation into the Central region to determine why strong sales are not translating into comparable profits, with particular attention to discounts, pricing, and product mix. Develop targeted marketing and sales initiatives to increase market penetration in the South region.

Sub-Category Performance Analysis

Row Labels	Sum of Sales		Sum of Profit	
Phones	\$	330,007.05	\$	44,515.73
Chairs	\$	328,449.10	\$	26,590.17
Storage	\$	223,843.61	\$	21,278.83
Tables	\$	206,965.53	\$	(17,725.48)
Binders	\$	203,412.73	\$	30,221.76

Machines	\$	189,238.63	\$	3,384.76
Accessories	\$	167,380.32	\$	41,936.64
Copiers	\$	149,528.03	\$	55,617.82
Bookcases	\$	114,880.00	\$	(3,472.56)
Appliances	\$	107,532.16	\$	18,138.01
Furnishings	\$	91,705.16	\$	13,059.14
Paper	\$	78,479.21	\$	34,053.57
Supplies	\$	46,673.54	\$	(1,189.10)
Art	\$	27,118.79	\$	6,527.79
Envelopes	\$	16,476.40	\$	6,964.18
Labels	\$	12,486.31	\$	5,546.25
Fasteners	\$	3,024.28	\$	949.52
Grand Total		2297200.86		286397.0217



Insights

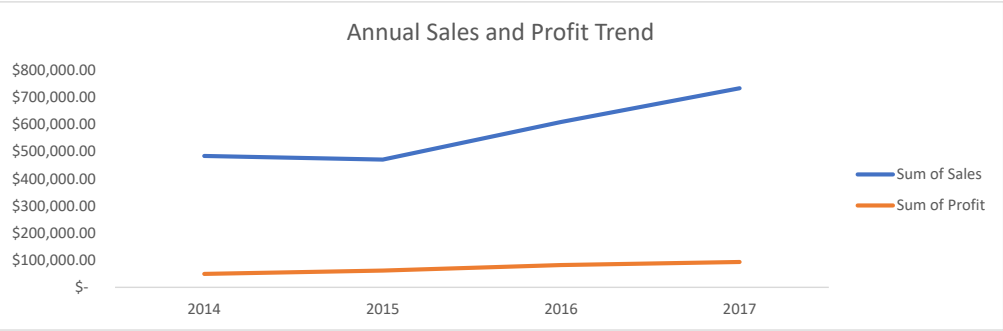
Phones generated the highest sales and strong profitability, making them the leading-performing sub-category. Chairs also contributed significantly to both sales and profit, indicating consistent customer demand. Tables generated substantial revenue but recorded a negative overall profit, making them one of the weakest-performing sub-categories. Bookcases produced relatively high sales but low profitability, suggesting reduced margins. Several sub-categories generated healthy profits despite lower sales, demonstrating that high revenue does not always translate into higher profitability.

Recommendation

Management should investigate the pricing strategy, discount levels, and cost structure of Tables and Bookcases to determine the causes of poor profitability. High-performing sub-categories such as Phones and Chairs should continue to receive strategic investment while maintaining their current profit margins.

SALES & PROFIT TREND ANALYSIS

Row Labels	Sum of Sales		Sum of Profit	
2014	\$	484,247.50	\$	49,543.97
2015	\$	470,532.51	\$	61,618.60
2016	\$	609,205.60	\$	81,795.17
2017	\$	733,215.26	\$	93,439.27
Grand Total		2297200.86		286397.0217



Insights

- Sales and profit showed a consistent upward trend from 2014 to 2017, indicating continuous business growth.
- The highest sales and profit were recorded in 2017, reflecting strong business expansion.
- No year experienced an overall decline in both sales and profit, suggesting stable business performance during the analysis period.

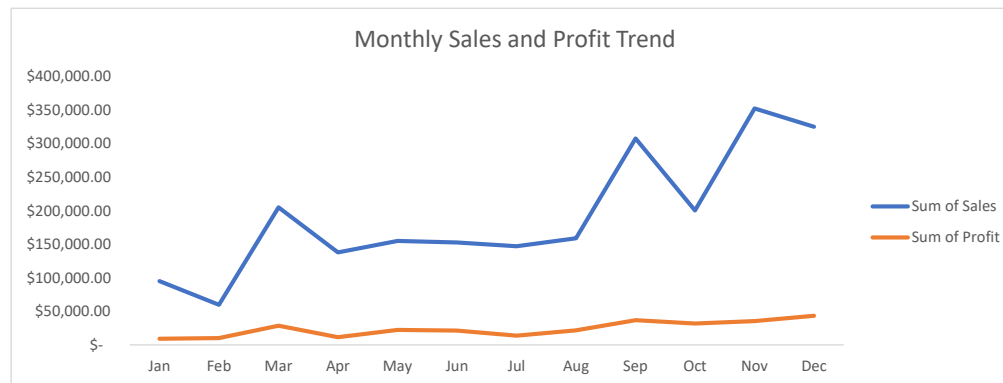
Recommendation

Continue implementing the business strategies that supported growth during the four-year period while monitoring profitability to ensure sales growth remains sustainable.

Monthly Trend

Row Labels	Sum of Sales		Sum of Profit	
Jan	\$	94,924.84	\$	9,134.45
Feb	\$	59,751.25	\$	10,294.61

Mar	\$	205,005.49	\$	28,594.69
Apr	\$	137,762.13	\$	11,587.44
May	\$	155,028.81	\$	22,411.31
Jun	\$	152,718.68	\$	21,285.80
Jul	\$	147,238.10	\$	13,832.66
Aug	\$	159,044.06	\$	21,776.94
Sep	\$	307,649.95	\$	36,857.48
Oct	\$	200,322.98	\$	31,784.04
Nov	\$	352,461.07	\$	35,468.43
Dec	\$	325,293.50	\$	43,369.19
Grand Total		2297200.86		286397.0217



Insights

- September recorded the highest sales, indicating strong customer demand during that month.
- November generated the highest profit despite having slightly lower sales than September, suggesting stronger profit margins.
- July recorded moderate sales but negative profitability, indicating possible excessive discounts or high operating costs.
- March achieved relatively strong sales but comparatively low profit.
- February recorded one of the lowest sales and profit values during the year.

Recommendation

Investigate the factors contributing to negative profitability in July and weaker margins in March. The strategies responsible for November's strong profitability should be examined and, where appropriate, replicated during other months.

ROOT CAUSE ANALYSIS

Investigation Question:

Why does the Furniture category generate relatively low profitability?

Answer:

Initial analysis showed that the Furniture category generated strong sales but comparatively low profitability. A deeper investigation revealed that the Tables and Bookcases sub-categories contributed significantly to these losses. Further product-level analysis identified several products with consistently negative profits despite high sales volumes. This suggests that pricing strategy, discounting, or cost structure may be reducing profitability within these product lines.

KEY FINDINGS

- Technology generated the highest sales and profit.
- Furniture generated high sales but relatively low profitability.
- Tables recorded the largest overall loss among all sub-categories.
- The West region outperformed all other regions.
- The Consumer segment generated the highest revenue.
- Sales and profit increased consistently from 2014–2017.
- November achieved the highest profitability, while July experienced negative profitability.

BUSINESS RECOMMENDATIONS

- Continue investing in high-performing Technology products.
- Review pricing and discount strategies for Tables and Bookcases.
- Investigate profitability challenges in the Central region.
- Expand successful sales strategies from the West region to other regions.
- Increase marketing efforts toward profitable customer segments.
- Investigate the causes of July's negative profitability.
- Monitor discounts to protect profit margins.

CONCLUSION

This exploratory data analysis identified the key factors influencing sales and profitability across products, customer segments, and regions. While the business demonstrates consistent growth and strong overall performance, opportunities exist to improve profitability by addressing underperforming product categories, optimizing discount strategies, and strengthening regional performance. The insights and recommendations provided in this report can support more informed, data-driven business decisions.